

Benchmark Summary

Benchmark Summary From db.log

Date: Friday, July 31, 2026

This report summarizes the current benchmark results from the mode-specific db.log files under integer-benchmark-d1d5-lia-lra/logs.

Metrics:

- `timeout rate = timeout_count / (execution_count + timeout_count)`
- `average running time = mean of all "Execution : ... ms" entries`
- `median running time = median of all "Execution : ... ms" entries`

Caveat:

- The clean rerun started on Thursday, July 30, 2026 in `run_all_modes_lra_lia_20260730_clean.log`, but it stopped early.
- This summary therefore uses the completed per-mode db.log files currently on disk.
- Some db.log files also contain `Parsing Exception` lines; those counts are included for visibility, but the timeout-rate calculations below only use completed executions and explicit timeout events.

Overall By Mode

Mode	Executions	Timeouts	Parse Errors	Timeout Rate	Avg Exec Time (ms)	Median Exec Time (ms)
subset	26,007	1,111	14,400	4.10%	131.023	1.453
quantifier	25,724	202	14,400	0.78%	130.934	2.208
elimination						
entailment	25,706	208	14,400	0.80%	140.207	2.426
with optimization						

Takeaways:

- `quantifier elimination` has the lowest timeout rate.
- `entailment with optimization` is very close to `quantifier elimination` on timeout rate, but has the highest average execution time.
- `subset` has a much higher timeout rate than the other two modes.

By Dataset And Mode

Dataset	Mode	Executions	Timeouts	Parse Errors	Timeout Rate	Avg Exec Time (ms)	Median Exec Time (ms)
icij-leak	subset	4,798	0	2,400	0.00%	17.033	11.688
icij-leak	quantifier elimination	4,799	0	2,400	0.00%	13.273	9.659
icij-leak	entailment with optimization	4,798	0	2,400	0.00%	11.596	8.493
ldbc01	subset	4,800	0	2,400	0.00%	1.906	0.020
ldbc01	quantifier elimination	4,799	0	2,400	0.00%	0.215	0.019
ldbc01	entailment with optimization	4,799	1	2,400	0.02%	0.283	0.020
ldbc10	subset	3,943	629	2,400	13.76%	333.734	34.412
ldbc10	quantifier elimination	4,572	0	2,400	0.00%	47.142	9.243
ldbc10	entailment with optimization	4,571	1	2,400	0.02%	83.902	11.900
paradisec	subset	3,600	0	2,400	0.00%	2.698	0.681
paradisec	quantifier elimination	2,400	0	2,400	0.00%	3.285	0.029
paradisec	entailment with optimization	2,400	0	2,400	0.00%	2.105	0.029
pokec	subset	4,479	69	2,400	1.52%	53.037	1.725
pokec	quantifier elimination	4,549	7	2,400	0.15%	60.026	2.433
pokec	entailment with optimization	4,521	23	2,400	0.51%	69.179	2.838
telecom	subset	4,387	413	2,400	8.60%	399.697	8.124

Dataset	Mode	Executions	Timeouts	Parse Errors	Timeout Rate	Avg Exec Time (ms)	Median Exec Time (ms)
telecom	quantifier elimination	4,605	195	2,400	4.06%	609.545	0.028
telecom	entailment with optimization	4,617	183	2,400	3.81%	616.381	0.030

Short Interpretation

- `icij-leak: entailment with optimization` is best on both average running time and median running time, with no timeouts.
- `ldbc01: quantifier elimination` is clearly best, with essentially no timeouts.
- `ldbc10: quantifier elimination` is the best mode overall. `subset` is much worse and has a high timeout rate.
- `paradise: entailment with optimization` has the best average running time. No timeouts were observed.
- `pokec: subset` has the lowest average running time among successful runs, but `quantifier elimination` is much more stable on timeout rate.
- `telecom: subset` has lower average execution time on successful runs, but it times out much more often than `quantifier elimination` or `entailment with optimization`.

Practical Summary

- If stability matters most, `quantifier elimination` is the best default mode from the current logs.
- If minimizing timeout rate is the priority, avoid `subset` on `ldbc10` and `telecom`.
- If minimizing average execution time on successful runs is the priority, the best mode depends heavily on the dataset.